# Sangbum Woo

*Woo Sangbum · [redacted]*

*PhD Student, Autonomous Driving & Robotics*

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## RESEARCH INTERESTS

End-to-End Autonomous Driving · SLAM & Localization · Path Planning and Decision Making · Real-vehicle Autonomous Systems and Robotics

## EDUCATION

2026 – present **Ph.D. in Automotive Engineering (AUTO Lab)**
*Hanyang University, Seoul, Republic of Korea*
Department of Automotive Engineering. Focus on end-to-end autonomous driving on real vehicles.

2023 – 2026 **M.S. in Computer Engineering, Future Convergence Engineering**
*Korea University of Technology and Education (KOREATECH)*
HPC Lab (SPIN). GPA 3.95 / 4.5.

2019 – 2023 **B.S. in Computer Science and Software**
*Korea University of Technology and Education (KOREATECH)*
Minor in Human Resource Development. GPA 3.31 / 4.5.

## RESEARCH EXPERIENCE

2023 – 2026 **Participating Researcher, Autonomous Driving SW Architecture**
*Korea Institute of Machinery & Materials (KIMM)*
Autonomous agricultural machinery for orchards. Contributed to SLAM-based mapping, Voronoi navigation, and the end-to-end ROS 2 autonomy stack on a real UGV platform.

- Developed topology-preserving incremental mapping for repetitive tree-row scenes.
- Designed Voronoi-based tree-row path planners for full-orchard coverage.
- Led architecture-level integration of perception, planning, and control modules on ROS 2.
- Conducted field experiments on a real UGV in working orchards.

2023 – 2026 **Graduate Research Assistant**
*KOREATECH HPC Lab → SPIN*
SLAM parallelization, multi-robot exploration, and autonomous driving systems research.

- Real-time map construction via parallelized SLAM pipelines.
- Space-partitioning-based exploration for multi-UGV teams.
- First-author submissions to IEEE Access and ICROS journals.

2021 – 2023 **President & Planning/Decision Team Lead**
*K-ROAD, KOREATECH Undergraduate Autonomous Vehicle Society*
Led the autonomous-driving undergraduate research society for three years, establishing its technical foundation and competition track record.

- Developed planning and decision algorithms for on-campus autonomous vehicles.
- Silver Prize, Creative Car Competition (2022); Hyundai AD Challenge Virtual Track (2023).
- Mentored junior members and oversaw long-term research planning.

## PUBLICATIONS

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|---|---|
| 2025 | **S. Woo**, C. Lee, and G. Jung. "Topology-Preserving Incremental Mapping and Voronoi-Based Tree-Row Navigation for Orchard Unmanned Ground Vehicles." *IEEE Access, under review (SCIE).* |
| 2023 | **S. Woo** and C. Lee. "Space Partitioning and Path Planning for Mission Area Exploration Using Multiple UGVs." *Journal of Institute of Control, Robotics and Systems (ICROS) (SCOPUS).* |

## PATENTS

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|---|---|
| Registered | **Real-Time Streaming Broadcast Chat Summarization System and Method** *Republic of Korea, Korean Intellectual Property Office (KIPRIS).* |

## AWARDS & HONORS

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| 2024 | **2nd Prize** (Korea Institute for Robot Industry Advancement President's Award) and **Special Prize**, Baemin Robot Delivery Challenge — Autonomous Driving Mission. *Woowa Brothers.* |
| 2022 | **Grand Prize**, Graduation Project Idea Competition. *KOREATECH.* |
| 2022 | **Best Paper Award**, Korea Software Congress (KSC). *Korean Institute of Information Scientists and Engineers (KIISE).* |
| 2022 | **Silver Prize**, Creative Car Competition — Autonomous Driving Division. *CARSA.* |

## TECHNICAL SKILLS

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| Languages | Python, C++, C, MATLAB, TypeScript, Bash. |
| Frameworks | ROS / ROS 2, PyTorch, OpenCV, Point Cloud Library (PCL), Open3D, TensorRT, ONNX, NumPy / SciPy. |
| Tools | Linux (Ubuntu), Git, Docker, CUDA, Gazebo, CARLA Simulator, RViz, CMake, Weights & Biases. |
| Domains | 3D object detection; LiDAR–camera sensor fusion; point-cloud processing; SLAM and localization; path, behavior, and motion planning; vehicle dynamics; PID / MPC control; trajectory tracking; Kalman filtering; deep learning; computer vision. |

## SELECTED PROJECTS

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| 2023 – 2025 | **Autonomous Driving for Orchard UGVs** Full stack for tree-row orchards: topology-preserving mapping, Voronoi row-navigation, and ROS 2 autonomy. With KIMM. |
| 2024 | **Baemin Robot Delivery Challenge** Outdoor last-mile delivery robot, autonomous-driving mission track. 2nd Prize + Special Prize. |
| 2023 | **Hyundai Autonomous Driving Challenge — Virtual Track** End-to-end autonomy stack in a high-fidelity urban simulator. |
| 2022 | **Creative Car Competition — Autonomous Driving** Self-built autonomous vehicle on a closed circuit. Silver Prize. |

## MEDIA COVERAGE & OUTREACH

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|---|---|
| 2024 | "Woowa Brothers Wraps Up Baemin Robot Delivery Challenge." *Robot News.* |
| 2023 | "KOREATECH Excels Across Robotics and Autonomous-Driving Competitions." *Nate News.* |
| 2023 | "KOREATECH Wins Grand and Gold Prizes at the 2023 Creative Mobility Competition." *Daejeon Today*. |
| 2023 | MBC public-relations feature on KOREATECH autonomous-driving research. *MBC.* |
| 2022 | "KOREATECH Wins Gold at the International Creative Car Competition." *Daehan Economic Daily*. |